

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

/ P Dinesh Reddy / 192525399 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	

Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

Title of AI Application

Generative AI Shopping Assistant: An AI-Based Personalized Retail Solution.

Domain Selection

Selected Domain: Retail

The retail industry is rapidly adopting Artificial Intelligence (AI) to improve customer experience and business performance. One of the latest AI applications in this field is the **Generative AI Shopping Assistant**.

These assistants use technologies such as **Natural Language Processing (NLP), Machine Learning (ML), Large Language Models (LLMs), Recommendation Systems, and Semantic Search** to deliver accurate and personalized shopping experiences.

Many leading e-commerce companies, including **Amazon, Walmart, Google Shopping, and Flipkart**, have started integrating AI shopping assistants into their platforms.

Introduction and Background

Artificial Intelligence (AI) has become one of the most important technologies in today's digital world. It enables computers and machines to perform tasks that normally require human intelligence, such as understanding language, learning from data, making decisions, and solving problems.

The retail sector has experienced significant growth with the rise of online shopping. Customers now have access to millions of products through e-commerce platforms. Traditional online shopping websites mainly rely on keyword-based searches, where customers must type exact product names or keywords.

To overcome these limitations, retailers are adopting **Generative AI Shopping Assistants**. These intelligent assistants allow customers to interact using natural language, just as they would with a human sales representative. Instead of searching with keywords, customers can ask complete questions such as:

- "Suggest the best smartphone under ₹25,000."
- "Which laptop is suitable for engineering students?"

- "Recommend comfortable running shoes for beginners."

The AI assistant understands the customer's request, searches the product database, compares different products, and provides personalized recommendations.

Generative AI Shopping Assistants use several AI technologies, including Machine Learning (ML), Natural Language Processing (NLP), Large Language Models (LLMs), and Recommendation Systems.

Many leading companies such as Amazon, Walmart, Google Shopping, and Flipkart have started using AI-powered shopping assistants to improve customer service and increase sales.

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Problem Identification

The rapid growth of online shopping has made it easier for customers to access a wide variety of products. However, the large number of available products also creates challenges when customers try to find the most suitable item..

Customers also face difficulty comparing similar products based on features, price, ratings, and reviews. Reading hundreds of product descriptions and customer reviews before making a purchase can be confusing. In addition, traditional customer support is limited by working hours and cannot always provide instant responses to customer queries.

Retail businesses also face challenges such as managing large product catalogs, understanding changing customer preferences, providing personalized recommendations, and maintaining customer satisfaction.

These challenges highlight the need for an AI-powered solution that can understand customer intent, provide accurate recommendations, and improve the overall shopping experience.

Need for AI Solution

Artificial Intelligence provides an effective solution to overcome the limitations of traditional online shopping systems. A **Generative AI Shopping Assistant** can understand customer queries written in natural language and provide personalized product recommendations based on individual needs and preferences.

The AI assistant helps customers by:

- Providing quick and accurate product recommendations.
- Understanding natural language questions instead of relying only on keywords.
- Comparing multiple products based on features, price, and customer reviews.
- Offering 24/7 customer support without human intervention.
- Reducing the time required to search for products.

For businesses, AI offers several advantages:

- Improves customer satisfaction through personalized shopping experiences.
- Increases sales by recommending relevant products.

- Reduces customer support workload through automation.
- Helps predict customer demand and manage inventory more efficiently.
- Enhances marketing by understanding customer preferences and shopping patterns.

By combining Machine Learning, Natural Language Processing, Recommendation Systems, and Search Algorithms, they provide faster and more accurate product recommendations while improving the experience for both customers and retailers. This also aligns with the course outcome of applying search algorithms and heuristic techniques to solve real-world search and optimization problems.

AI Technologies Used

A **Generative AI Shopping Assistant** combines several Artificial Intelligence technologies to understand customer queries, search for suitable products, and provide personalized recommendations. Each technology plays an important role in improving the shopping experience.

AI Technologies Used

AI Technology	Purpose in Shopping Assistant
Machine Learning (ML)	Learns customer preferences from purchase history and browsing behavior to improve recommendations.
Natural Language Processing (NLP)	Understands customer questions written in natural language and identifies their intent.
Large Language Models (LLMs)	Generates human-like responses, explains product features, and answers customer queries.
Recommendation System	Suggests products based on customer interests, ratings, and previous purchases.
Semantic Search	Searches products based on the meaning of the query instead of exact keywords.
Generative AI	Creates personalized shopping suggestions, product comparisons, and buying advice.

Explanation of AI Technologies

1. Machine Learning (ML)

Machine Learning helps the shopping assistant learn from customer behavior. It studies purchase history, search history, product ratings, and browsing patterns to recommend products that match the customer's interests. As more data is collected, the recommendations become more accurate.

2. Natural Language Processing (NLP)

Natural Language Processing enables the AI to understand customer questions written in everyday language. For example, if a customer asks, "**Suggest the best laptop under ₹60,000,**" the AI identifies the product category (laptop), the budget (₹60,000), and the user's intent (recommendation).

3. Large Language Models (LLMs)

Large Language Models generate detailed and human-like responses. They explain product specifications, compare different products, and answer follow-up questions, making the interaction more natural and informative.

4. Recommendation System

The recommendation system analyzes customer preferences, previous purchases, product ratings, and shopping trends. It then suggests products that are most relevant to the customer, increasing the chances of a successful purchase.

5. Semantic Search

Unlike traditional keyword search, semantic search understands the meaning behind the customer's query. For example, if a customer searches for "**comfortable shoes for walking,**" the AI recommends suitable walking shoes even if the exact words are not present in the product title.

6. Generative AI

Generative AI creates personalized responses instead of displaying only a list of products. It can compare products, explain features, suggest alternatives, and provide buying advice in a conversational manner.

Role of AI Technologies in the Shopping Assistant

The integration of these AI technologies enables the shopping assistant to:

- Understand customer requirements accurately.
- Provide personalized product recommendations.
- Improve product search results.
- Reduce customer search time.
- Enhance customer satisfaction through intelligent conversations.

Together, these technologies make online shopping faster, easier, and more efficient for both customers and retailers.

Working Mechanism

A **Generative AI Shopping Assistant** works by understanding the customer's query, searching for relevant products, and providing personalized recommendations. It combines Natural Language Processing (NLP), Machine Learning (ML), Large Language Models (LLMs), and Recommendation Systems to deliver accurate and helpful responses.

The complete working process is explained below.

Step 1: Customer Enters a Query

The customer asks a question using natural language.

Examples:

- "Suggest the best smartphone under ₹25,000."
- "Which laptop is best for engineering students?"
- "Recommend running shoes for beginners."

The AI understands the customer's request instead of simply searching for keywords.

Step 2: Query Processing

The Natural Language Processing (NLP) module analyzes the customer's question and identifies important information such as:

- Product category
- Budget
- Brand preference
- Required features
- Customer intent

Step 3: Product Search

The AI searches the product database using **semantic search**, which understands the meaning of the customer's query. This provides more relevant results than traditional keyword-based searches.

Step 4: Product Ranking

The Recommendation System compares products using factors such as:

- Customer ratings
- Product reviews
- Price
- Popularity
- Previous purchase history
- Customer preferences

The products that best match the customer's requirements are ranked higher.

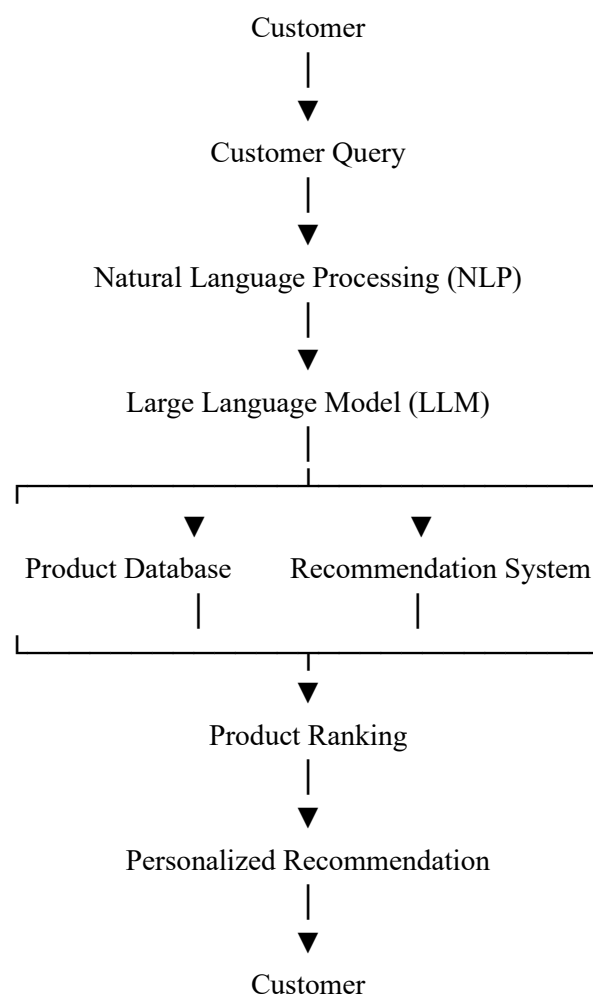
Step 5: AI Response Generation

The Large Language Model (LLM) generates a personalized response. It may compare products, explain their features, and recommend the best option based on the customer's needs.

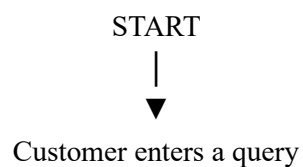
Step 6: Customer Feedback

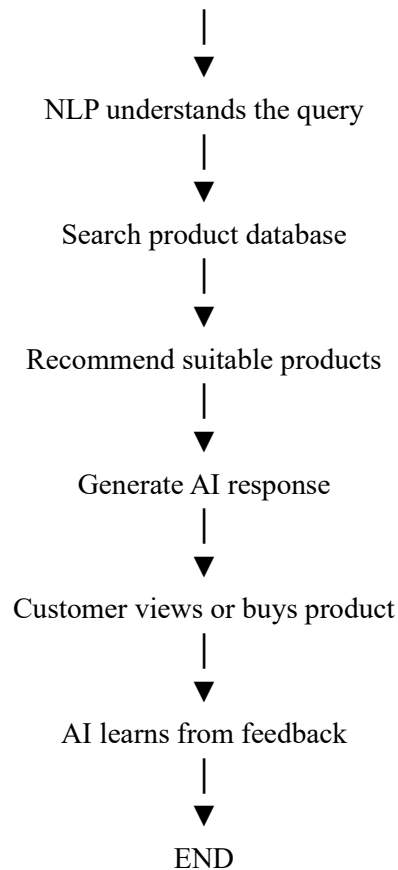
The AI learns from customer actions, such as clicks, purchases, ratings, and reviews. This helps improve future recommendations.

System Architecture Diagram



Working Flowchart





search Algorithms and Heuristic Techniques (CO1)

This application uses **search algorithms** and **heuristic techniques** to quickly identify the most suitable products for customers, supporting the course outcome on search and optimization.

Search Algorithms

Search algorithms help the AI find relevant products from a large product database. Instead of matching only keywords, the AI uses **semantic search** to understand the meaning of the customer's query and retrieve the most appropriate products.

Heuristic Techniques

A heuristic is a method used to find a good solution quickly without checking every possible option. The shopping assistant ranks products using factors such as:

- Product relevance
- Customer ratings
- Price
- Reviews
- Popularity

This ranking helps the AI recommend the best products in less time, improving customer satisfaction and shopping efficiency.

Data Sources and Processing

A Generative AI Shopping Assistant requires different types of data to provide accurate and personalized product recommendations. The quality of the AI's response depends on the quality of the data it receives. The system collects, processes, and analyzes data before generating recommendations.

Data Sources

The shopping assistant collects information from multiple sources, including:

1. Customer Data

This includes:

- Purchase history
- Search history
- Browsing history
- Wishlist
- Shopping cart
- Customer preferences

This information helps the AI understand customer interests and recommend suitable products.

2. Product Data

The product database contains:

- Product name
- Category
- Brand
- Price
- Features
- Images
- Stock availability

The AI uses this information to compare products and provide accurate recommendations.

3. Customer Reviews and Ratings

Customer feedback helps the AI identify popular and highly rated products. Products with better ratings and positive reviews are more likely to be recommended.

4. Business Data

Retail companies also provide:

- Inventory details
- Discount offers
- Sales records
- Delivery information

This helps the AI recommend products that are available and offer good value.

Data Processing

Before generating recommendations, the AI processes the collected data through several stages.

Step 1: Data Collection

The AI gathers data from customer interactions, product catalogs, reviews, and business databases.

Step 2: Data Cleaning

Incorrect, duplicate, and outdated information is removed to improve data quality.

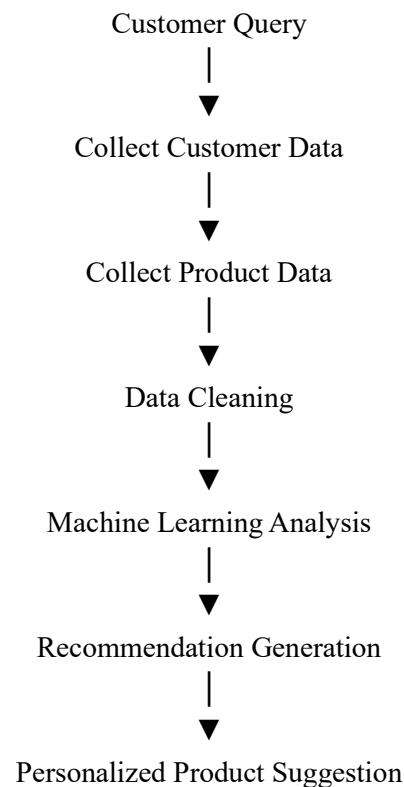
Step 3: Data Analysis

The AI analyzes customer preferences, shopping history, product ratings, and prices using Machine Learning algorithms.

Step 4: Recommendation Generation

Based on the analysis, the AI identifies the most suitable products and generates personalized recommendations for the customer.

Data Processing Flow Diagram



Importance of Data Processing

Proper data processing helps the shopping assistant:

- Provide accurate recommendations.
- Improve search results.
- Reduce incorrect suggestions.
- Understand customer preferences.
- Increase customer satisfaction.

Real-World Implementation

Generative AI Shopping Assistants are already being used by many leading retail companies to improve customer experience and increase sales. These AI systems help customers find products quickly, answer shopping-related questions, and provide personalized recommendations.

1. Amazon – Rufus AI Shopping Assistant

Amazon introduced **Rufus**, an AI-powered shopping assistant that helps customers search for products using natural language. Customers can ask questions like:

- "Which laptop is best for students?"
- "Compare iPhone and Samsung smartphones."

Benefits:

- Faster product search
- Personalized recommendations
- Better shopping experience
- Improved customer satisfaction

2. Flipkart AI Assistant

Flipkart uses AI technologies to improve product search and customer support. The AI assistant helps customers find products quickly, recommends similar products, and answers common questions related to orders and shopping.

Benefits:

- Easy product discovery
- Personalized recommendations
- Faster customer service
- Improved user experience

Comparison of AI Shopping Assistants

Company AI Application Main Features

Amazon Rufus AI Product recommendations, comparisons, conversational shopping

Flipkart AI Assistant Smart search, recommendations, customer support

Impact and Benefits

Generative AI Shopping Assistants make online shopping easier and faster. They help customers find the right products by understanding their needs and giving personalized recommendations. Customers can also compare products and get instant answers to their questions.

For businesses, AI helps increase sales by suggesting products that customers may like. It also reduces the workload of customer support teams by answering common questions automatically. In addition, AI helps businesses understand customer preferences and manage products more efficiently.

Overall, Generative AI Shopping Assistants improve the shopping experience for customers and help businesses work more effectively.

Benefits of Generative AI Shopping Assistant

Benefits for Customers

Faster product search

Personalized recommendations

Easy product comparison

24/7 assistance

Better shopping experience

Benefits for Businesses

Increased sales

Better inventory management

Reduced customer support workload

Improved business efficiency

Higher customer satisfaction

Challenges and Limitations

Generative AI Shopping Assistants are useful, but they also have some challenges.

Challenges

- AI needs correct and updated product information to give accurate recommendations.
- Customer data must be kept safe and secure.
- Sometimes AI may give incorrect answers or suggestions.
- Setting up AI systems can be expensive.

Limitations

- AI requires an internet connection to work.
- It may not understand every customer query correctly.
- Human support is still needed for difficult problems.

Future Scope and Emerging Trends

The future of Generative AI in retail is very promising. AI shopping assistants will become smarter and provide better services to customers.

Some future improvements are:

- **Voice Shopping:** Customers can search and buy products using voice commands.
- **Better Recommendations:** AI will suggest products more accurately based on customer interests.
- **Virtual Try-On:** Customers can try clothes, shoes, or glasses online before buying.
- **Smart Stores:** AI and robots may help customers in retail stores.
- **Faster Shopping:** AI will make product search and recommendations quicker and more accurate.

In the future, Generative AI Shopping Assistants will make online shopping easier, faster, and more personalized.

Conclusion

Generative AI Shopping Assistants have made online shopping easier and more convenient. They help customers find the right products, compare different options, and receive personalized recommendations. By using technologies like **Machine Learning (ML)**, **Natural Language Processing (NLP)**, and **Large Language Models (LLMs)**, the shopping assistant understands customer needs and provides accurate suggestions.

These AI systems benefit both customers and businesses. Customers save time and enjoy a better shopping experience, while businesses can increase sales, improve customer satisfaction, and reduce customer support work.

The shopping assistant also uses **search algorithms** and **heuristic techniques** to search for and recommend the best products quickly, supporting the course outcome of this assessment.

In the future, Generative AI Shopping Assistants will become more advanced and continue to improve the online shopping experience.

References

1. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th Edition, Pearson, 2021.
2. OpenAI, *GPT Models and API Documentation*, 2025.
3. Amazon Science, *Amazon Rufus AI Shopping Assistant*, 2025.
4. Google Cloud, *Generative AI for Retail*, 2025.
5. Microsoft, *Azure AI for Retail*, 2025.
6. IEEE Xplore Digital Library, *Research Papers on Generative AI in Retail*, 2024–2025.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.